

Câu 1:

$$D = \begin{vmatrix} 1 & 1 & 2 & -2 \\ 2 & 3 & 1 & 4 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 1 & 2 \end{vmatrix}$$

$$\det D = 1 \cdot A_{11} + 2 \cdot A_{21} + 0 \cdot A_{31} + 0 \cdot A_{41}$$

$$A_{11} = (-1)^2 \begin{vmatrix} 3 & 1 & 4 \\ 1 & 2 & -1 \\ 1 & 1 & 2 \end{vmatrix} = 8$$

$$A_{21} = (-1)^3 \begin{vmatrix} 1 & 2 & -2 \\ 1 & 2 & -1 \\ 1 & 1 & 2 \end{vmatrix} = -1$$

$$\det D = 8 - 2 = 6$$

Câu 4:

$$X \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} X = \begin{pmatrix} 4 & 5 \\ -9 & 0 \end{pmatrix}$$

Giải
Ta giả thiết to suy ra X có cỡ 2×2 giả $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$P \Leftrightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ -9 & 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} 3a+2b & a+2b \\ 3c+2d & c+2d \end{pmatrix} - \begin{pmatrix} a+2c & b+2d \\ 3a+4c & 3b+4d \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ -9 & 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2a + 2b - 2c & c + b - 2d \\ 3c + 2d - 3a - 4c & c - 3b - 2d \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ -9 & 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 2a - 2b - 2c = 4 \\ a + b - 2d = 5 \\ -3a - c + 2d = -9 \\ -3b + c - 2d = 0 \end{cases} \Rightarrow \begin{cases} a = 2 \\ b = 1 \\ c = 1 \\ d = -1 \end{cases}$$

$$\Rightarrow X = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$$